

MALIHA ZAHAN CHOWDHURY

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RESEARCH INTEREST

Large Language Models, Federated Learning, Machine Learning, Natural Language Processing, Computer Vision.

EDUCATION

Northern Illinois University

Ph.D. Computer Science; CGPA: 3.89/4.00

Dekalb, IL, USA

August 2025 - present

Chittagong University of Engineering and Technology (CUET)

B.Sc. in Computer Science and Engineering; CGPA: 3.63/4.00

Chittagong, Bangladesh

February 2017 - September 2022

COURSEWORK

Ongoing Courses: Pattern Recognition & Data Mining II, Topics in Computer Science, Grad Reading Computer Science

Completed Courses: Programming Principle in Python, Big Ideas in Computer Science, Pattern Recognition & Data Mining I, Applied Machine Learning

WORK EXPERIENCE

Northern Illinois University

Research Assistant, Dr. Zhishuai Guo's Lab

Dekalb, IL

January 2026 - Present

- Assisting in the development of an ongoing project involving skeleton data and federated learning.

Northern Illinois University

Teaching Assistant, CSCI 467 Intro to Software Engineering.

Dekalb, IL

August 2025 - December 2025

- Assessed and provided constructive feedback on the final project, focusing on the development of responsive websites.
- Offered weekly office hours and additional one-on-one tutoring sessions, aiding students in overcoming technical challenges and deepening their understanding of software development process.

East Delta University (EDU)

Lecturer, School of Science, Engineering and Technology

Chittagong, Bangladesh

October 2024 to Present

- Courses Instruct:** Artificial Intelligence, Object-Oriented Programming
- Other Responsibilities:** Co-advisor of EDU Debating Club

Port City International University (PCIU)

Lecturer, Department of Computer Science and Engineering

Chittagong, Bangladesh

November 2022 to September 2024

- Courses Instructed:** Pattern Recognition, Operating System Concepts, Structured Programming Language.
- Other Responsibilities:**
 - Undergraduate thesis supervision
 - Advisor of a running batch
 - Meeting minutes committee Convener.
 - Class routine and thesis/project committee member.

E-soft Limited

Industrial Attachment

Dhaka, Bangladesh

February 2022

BACHELOR THESIS

A blockchain based decentralized document authentication System for multiple organizations

- Supervisor:** Dr. Asaduzzaman, Professor, Dept. of CSE, CUET.
- This project was created to verify the authenticity of crucial documents for various companies. The Interplanetary File System (IPFS) and the Ethereum blockchain are used in the endeavor to create a safe environment, as both of these structures make the system decentralized. All three panels in the system need transactions on the blockchain to confirm a decision. The system is further secured by the requirement that each organization face an admin verification procedure to make an individual an admin by paying a pre-required amount of Ether through a blockchain transaction.
- Tools:** Solidity, Javascript, React.js, VS studio, JSON, InterPlanetary File System (IPFS).

PUBLICATIONS

- [1] **Bengali Hate Speech Categorization: A Comparison of Hybrid Deep Learning and Transformer-Based Models** [Paper]
I. Jahan, M. Z. Chowdhury, S. Nur, Z. R. Chowdhury, R. Barua, and T. Amrine
2024 International Conference on Recent Progresses in Science, Engineering and Technology (In Press) [ICRPSET 2024]
- [2] **A Blockchain-Based Decentralized Document Authentication System for Multiple Organizations.** [Paper] [Link]
M. Z. Chowdhury and Asaduzzaman
IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering [WIECON-ECE 2022]

TECHNICAL SKILLS

Programming Languages: C, C++, PYTHON, JAVA, PHP.

Web Technologies: HTML, CSS, React.js.

Content Writing: LaTeX & Microsoft Office.

ML/NLP Libraries: NLTK, BNLTK, BLTK, scikit-learn, tensorflow, matplotlib, pandas, numpy.

PROJECTS

Multimodal Hierarchical Classification of Biomedical Research Figures | Fall 2025

- Conducted a hierarchical classification on biomedical figures utilizing a subset of the BIOMEDICA dataset, featuring 15 coarse classes at Level 1 and 82 fine-grained classes at Level 2.
- Established a vision-only baseline for both classification levels.
- Offered empirical evidence that captions are essential for addressing visual ambiguities in fine-grained categories.

Enhancing Emotion and Sentiment Recognition in Multimodal Conversations: Addressing Class Imbalance in the MELD Dataset | Fall 2025

- Effectively employed dialogue-level oversampling and square-root scaled class weights to address class imbalance in the MELD dataset.
- Showed notable improvements in weighted F1-scores for sentiment and emotion classification, particularly in text and bimodal modalities.

Attendance App [GitHub]

- An Android app designed to help teachers record student attendance in any class or course, and it calculates the final attendance percentage as well.
- **Language:** Java, **Database:** MySQL.

Plagiarism Detection of Student Submitted Code [GitHub]

- A web application that examines students' submitted code to determine the plagiarism percentage and stores the related information.
- **Language:** Java.

Picturesque (App + Website) [GitHub]

- This project is a hybrid app and website designed for hiring photographers based on user preferences. Users can select the occasion they need a photographer for and explore photographer profiles to find one that matches their criteria.
- **Language:** PHP, HTML, CSS, **Database:** MySQL, **Framework:** React Native.

PROFESSIONAL TRAINING

- **Faculty Training and Development Program** [Certificate]
Organized by the Institutional Quality Assurance Cell (IQAC), Port City International University, Bangladesh.
- **Concept and Development of Outcome Based Education (OBE) Curriculum for Quality Improvement in Higher Education** [Certificate]
Organized by the Institutional Quality Assurance Cell (IQAC), Port City International University, Bangladesh.

CO-CURRICULAR ACTIVITIES

- Paper presentation at ICNST 2019. **(Best Presenter Award)** [Certificate]
Title: Plagiarism Detection of Student's Submitted Code
- Vice President at CUET Debating Society | June 2020 - August 2021.
- Participated in 12th Inter-Department Bangla Debate and English Parliamentary Debate Competition organized by CUET Debating Society | 2017. [Certificate] [Certificate]
- Judges Coordinator, HULT PRIZE CUET (2019 - 2020) and Director of Judges Coordinator at HULT PRIZE CUET (2020 - 2021). [Certificate] [Certificate]
- Participated at HULT PRIZE CUET as a Competitor | 2020 - 2021. [Certificate]

REFERENCES

Dr. Zhishuai Guo

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Dr. Asaduzzaman

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